



CSE470: Software Engineering

SET - A

Semester: **Summer 2025**
Examination: **Quiz 4**

Time: **25 minutes**
Full marks: **15**

Name: _____

ID: _____

Section: **8**

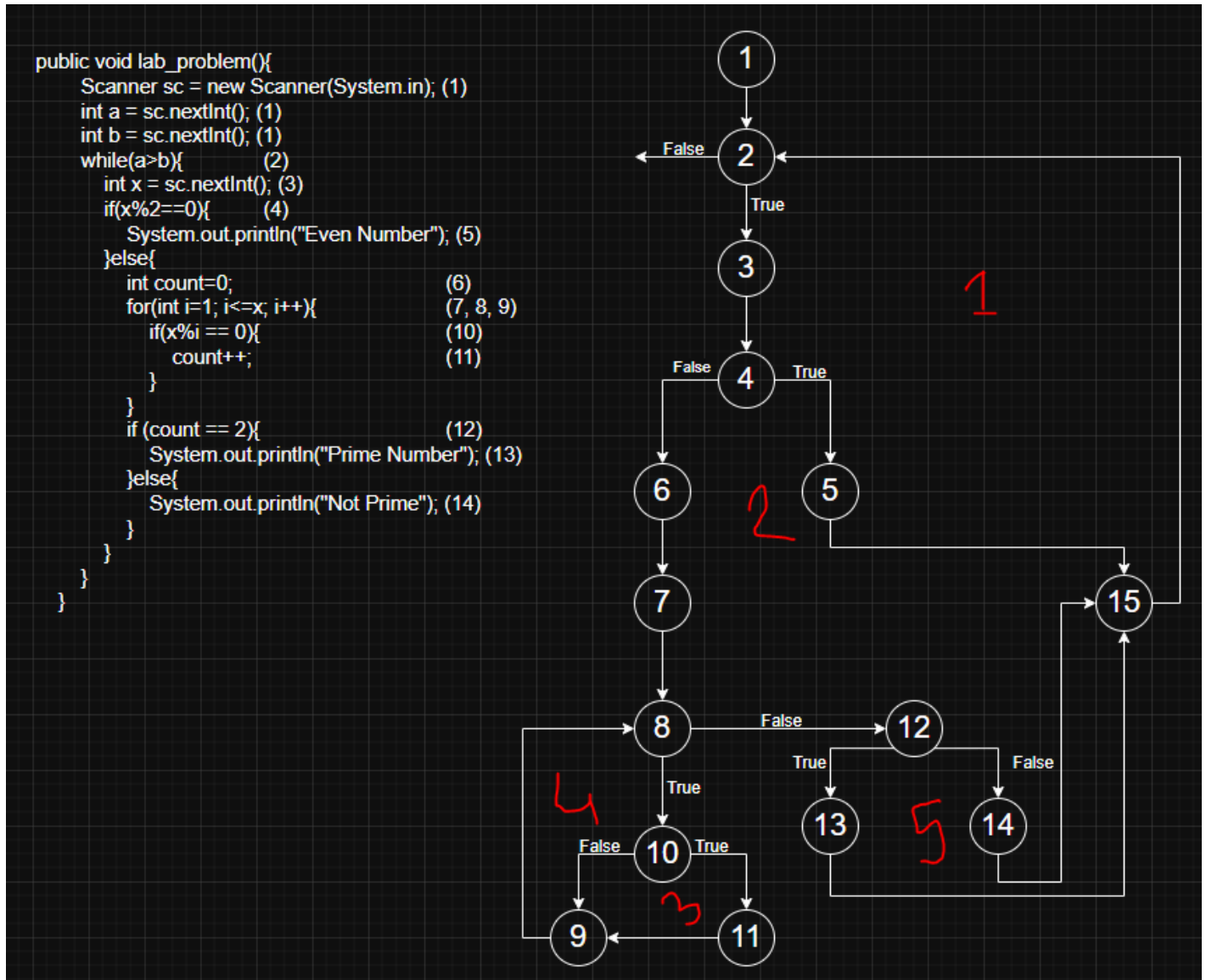
Read the code and answer the following questions:

Python	Java
<pre>class Human: def eat(self): print("Human Eats") def sleep(self): print("Human Sleeps") class BRACUStudent(Human): def verify(self): print("Student Verifies") def study(self): print("Student Studies") class CSEStudent(BRACUStudent): def study(self): print("Student studies CSE") def lab_problem(self): a = input("Enter a number") b = input("Enter another number") while(a>b): x = input("Enter a number") if (x%2 == 0): print("Even Number") else: count=0 for i in range(1,x+1): if(x%i == 0): count+=1 if count == 2: print("Prime Number") else: print("Not Prime")</pre>	<pre>public class Human { public void eat(){ System.out.println("Human Eats"); } public void sleep(){ System.out.println("Human Sleeps"); } } public class BRACUStudent extends Human{ public void verify(){ System.out.println("Student Verifies"); } public void study(){ System.out.println("Student Studies"); } } public class CSEStudent extends BRACUStudent{ public void study(){ System.out.println("Student studies CSE"); } public void lab_problem(){ Scanner sc = new Scanner(System.in); int a = sc.nextInt(); int b = sc.nextInt(); while(a>b){ int x = sc.nextInt(); if(x%2==0){ System.out.println("Even Number"); }else{ int count=0; for(int i=1; i<=x; i++){ if(x%i == 0){ count++; } } if (count == 2){ System.out.println("Prime Number"); }else{ System.out.println("Not Prime"); } } } } }</pre>

A. Answer the following questions for the “lab_problem” method:

1. Write the nodes on the question paper and draw the CFG [1+4]
2. Find out all the independent paths in your CFG [4]
3. Calculate the Cyclomatic Complexity [3]
4. Write a testcase to recreate a path that has greater than or equal to 5 nodes [1]

B. Calculate Specialization Index (SIX) for the CSEStudent class [2]



Complexity = 5+1 = 5+1 = 19-15+2 = 6

SIX:

Human → BRACUStudent → CSEStudent

DIT = 2, NMA (lab_problem) = 1, NMI (verify, eat, sleep) = 3, NMO (study) = 1

$SIX = (2 \times 1) / (1 + 3 + 1) \times 100\% = 40\%$



BRAC University
Department of Computer Science and Engineering (CSE)

CSE470: Software Engineering

SET - B

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Name: _____

ID: _____

Section: **8**

Read the code and answer the following questions:

Python	Java
<pre> class SultansDine: def kacchi(self): print("Sells Kacchi") def tehari(self): print("Sells Tehari") def morog_polao(self): print("Sells Morog polao") def tough_calculation(self): a = input("Enter a number") b = input("Enter another number") while(a>b): x = input("Enter a number") if (x%2 == 0): count = 0 for i in range(1, x): if (i*i == x): count+=1 if count == 1: print("Perfect Square") else: print("Not Perfect Square") else: print("Odd Number") class Banani(SultansDine): def tehari(self): print("Does not sell Tehari") def wedding_platter(self): print("Sells Wedding Platter") class Gulshan(SultansDine): def kacchi(self): print("Does not sell Kacchi") def chaap(self): print("Sells Chaap") </pre>	<pre> class SultansDine { public void kacchi() { System.out.println("Sells Kacchi"); } public void tehari() { System.out.println("Sells Tehari"); } public void morog_polao() { System.out.println("Sells Morog Polao"); } public void tough_calculation() { Scanner sc = new Scanner(System.in); int a = sc.nextInt(); int b = sc.nextInt(); while (a > b) { int x = sc.nextInt(); if (x % 2 == 0) { int count = 0; for (int i = 1; i < x; i++) { if (i * i == x) { count++; } } if (count == 1) { System.out.println("Perfect Square"); } else { System.out.println("Not Perfect Square"); } } else { System.out.println("Odd Number"); } } } } public class Banani extends SultansDine{ public void tehari(){ System.out.println("Does not sell Tehari"); } public void wedding_platter(){ System.out.println("Sells Wedding Platter"); } } </pre>

```
}

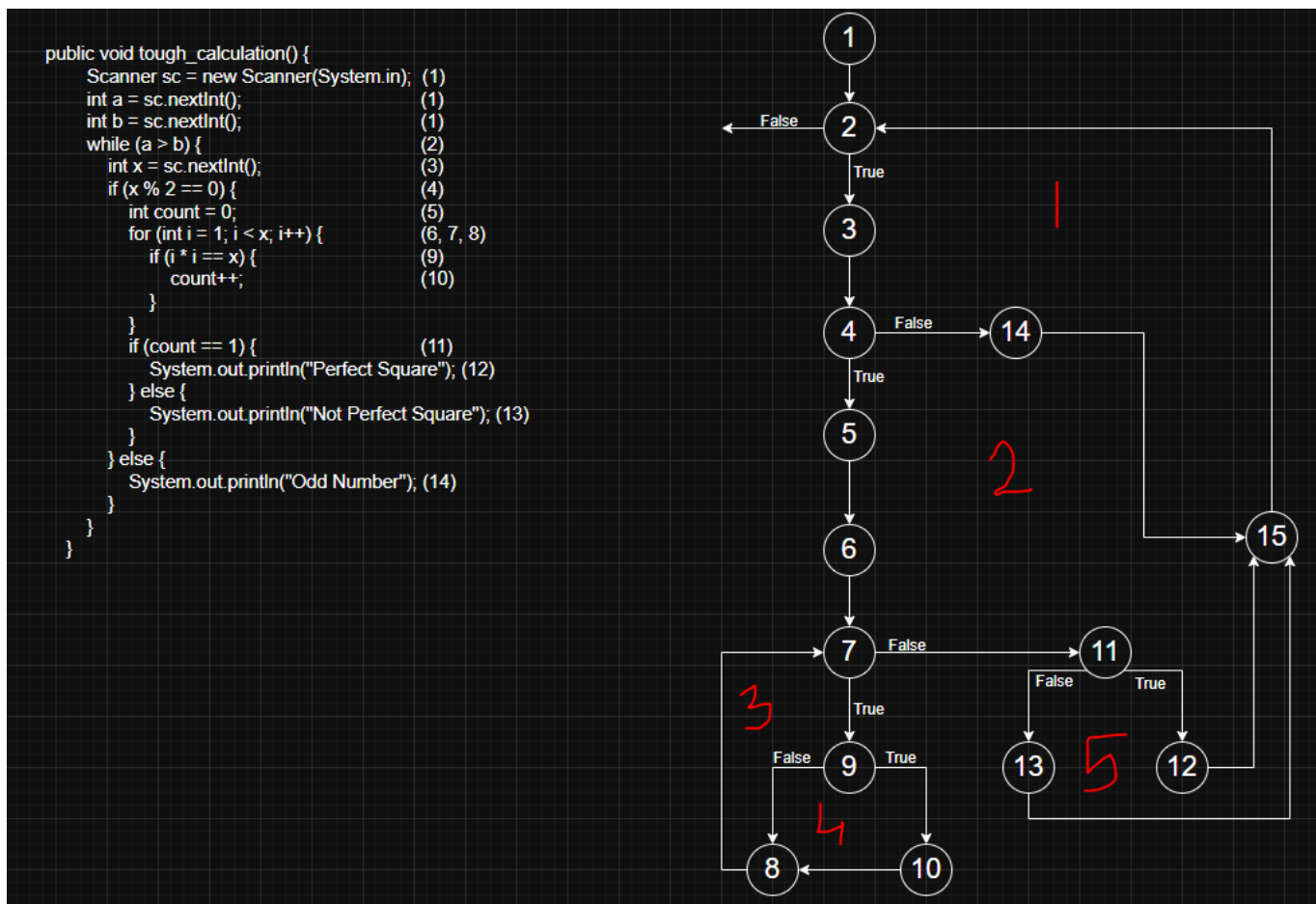
```

```
public class Gulshan extends SultansDine{
    public void kacchi(){
        System.out.println("Does not sell kacchi");
    }
    public void chaap(){
        System.out.println("Sells Chaap");
    }
}
```

A. Answer the following questions for the “tough_calculation” method:

1. Write the nodes on the question paper and draw the CFG [1+4]
2. Find out all the independent paths in your CFG [4]
3. Calculate the Cyclomatic Complexity [3]
4. Write a testcase to recreate a path that has greater than or equal to 5 nodes [1]

B. Calculate Specialization Index (SIX) for the Gulshan class [2]



Complexity = 5+1 = 5+1 = 19-15+2 = 6

SIX: SultansDine → Banani, SultansDine → Gulshan

DIT = 1, NMA (chaap) = 1, NMI (tehari, morog_polao, tough_calc) = 3, NMO (kacchi) = 1

SIX = $(1 \times 1) / (1 + 3 + 1) \times 100\% = 20\%$